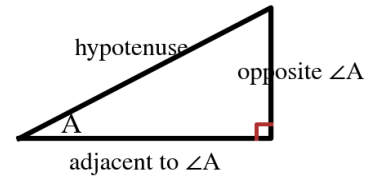


Speak Like A Mathematician:

trigonometric ratio:

sine · cosine · tangent:

opposite · adjacent (relative to an angle):



SOH - CAH - TOA

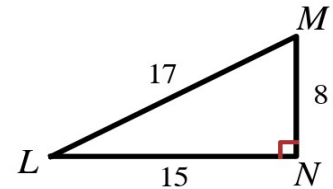
$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$ · $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$ · $\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$

To find a missing side: 1) label the sides opp / adj / hyp from the given angle, 2) pick the ratio that uses the side you _____ and the side you _____, 3) set up the equation and solve.

Example 1 - Write the Six Ratios

In $\triangle LMN$ at the right, the right angle is at N, $LN = 15$, $NM = 8$, and $LM = 17$. Write each ratio as a fraction and a four-place decimal.

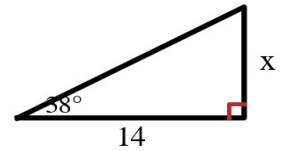
- $\sin L$, $\cos L$, $\tan L$
- $\sin M$, $\cos M$, $\tan M$
- What do you notice about $\sin L$ and $\cos M$?



Example 2 - Find a Missing Side (worked)

In the triangle at the right, the 38° angle has adjacent side 14 and opposite side x .

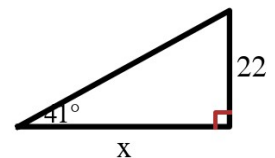
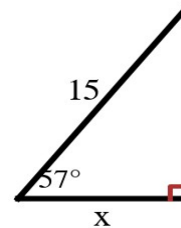
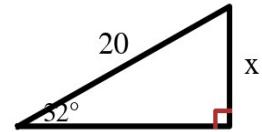
- Which ratio applies, and why?
- Set up and solve. Round to the nearest tenth.



Example 3 - Find the Missing Side

Round each answer to the nearest tenth. Figures at the right.

- 32° angle, hypotenuse 20, opposite x .
- 57° angle, hypotenuse 15, adjacent x .
- 41° angle, opposite 22, adjacent x .



Example 4 - A Preview of Inverse Trig

You stand 275 ft from a 600-ft skyscraper (figure at the right).

- Write the tangent of the angle of elevation x° .
- Find x with the $\tan^{-1}$ key. Round to the nearest tenth.

